

# Technical Report: Clinical Text Codification

ICD-10 First-Character Chapter Classification NLP Challenge

## NLP Group 16

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### Abstract

Automating clinical codification is a critical task in healthcare administration, traditionally requiring manual translation of free-text clinical literals into structured codes. This report presents an end-to-end evaluation of a deep learning approach for predicting the first alphanumeric character of ICD-10 codes from short medical text literals in Spanish. We describe the exploratory data analysis of 13,700 samples, detail the construction of a TF-IDF and Logistic Regression baseline, and compare it against a deep learning architecture using a pre-trained clinical Spanish transformer (`roberta-base-biomedical-clinical-es`). The deep learning model achieves an accuracy of **57.01%** and a Macro- $F_1$  score of **49.42%**, representing absolute improvements of **+4.02%** and **+4.93%** over the baseline, respectively. Finally, we conduct a qualitative error analysis to characterize failure modes, including semantic overlaps, looking-glass character confusions, and dataset annotation noise.

## 1 Introduction

Clinical codification involves mapping free-text medical records (diagnoses, symptoms, and procedures) to standardized alphanumeric codes defined by the International Classification of Diseases, 10th Revision (ICD-10). In this challenge, the goal is scoped to predicting the first alphanumeric character of the target ICD-10 code based on a short clinical literal (e.g., “*golpe contra muro*”, “*itu pARTO*”). The first character indicates the broad medical category or chapter:

- **Letters (A–Z):** Diagnosis chapters (e.g., A and B for infectious diseases, C for malignant neoplasms, Z for factors influencing health status).
- **Numbers (0–9):** Medical, surgical, and obstetrical procedure chapters.

The prediction task is structured as a 36-class single-label classification problem. The data is split into an 80% training set and a 20% validation set using a stratified split to preserve class proportions.

## 2 Exploratory Data Analysis (EDA)

We performed exploratory data analysis on the dataset using the script `eda_analysis.py`.

### 2.1 Basic Corpus Statistics

The dataset contains a total of **13,700 clinical text literals** spanning all 36 possible categories. Standard text length analysis shows that the literals are very short:

- **Character Length:** Mean of 16.9 characters (standard deviation  $\sigma = 10.3$ ).
- **Word Count:** Mean of 2.2 words (standard deviation  $\sigma = 1.4$ ), with a median of 2.0 words, and a maximum length of 18.0 words.

## 2.2 Severe Class Imbalance

The dataset is characterized by an extreme class imbalance, which presents a significant challenge for model training and evaluation. The class counts range from a maximum of 1,715 samples to a minimum of 7 samples, yielding an **imbalance ratio of 245.0x**.

- **Top 3 Most Frequent Categories:**

1. Z (Health status / Services): 1,715 samples (12.52% of the corpus)
2. O (Pregnancy / Childbirth): 1,507 samples (11.00%)
3. O (Procedures - Medical/Surgical): 1,141 samples (8.33%)

- **Least Frequent Categories (Support  $\leq 20$ ):**

- W (External causes - falls): 7 samples (0.05%)
- X (External causes - other): 11 samples (0.08%)
- A (Infectious diseases - bacteria/viruses): 18 samples (0.13%)
- Y (External causes - surgery/medical): 36 samples (0.26%)
- U (Special purposes): 21 samples (0.15%)

The class frequency and text length distributions are visualized in Figure 1 and Figure 2, respectively.

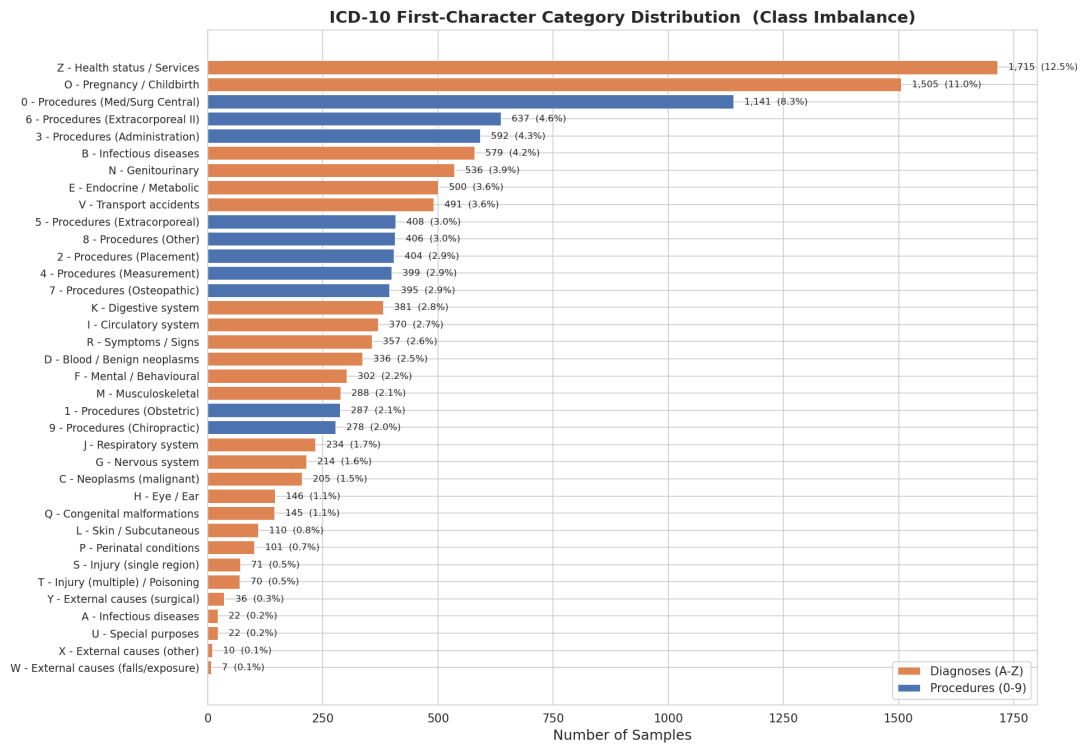


Figure 1: Class Frequencies across Chapters

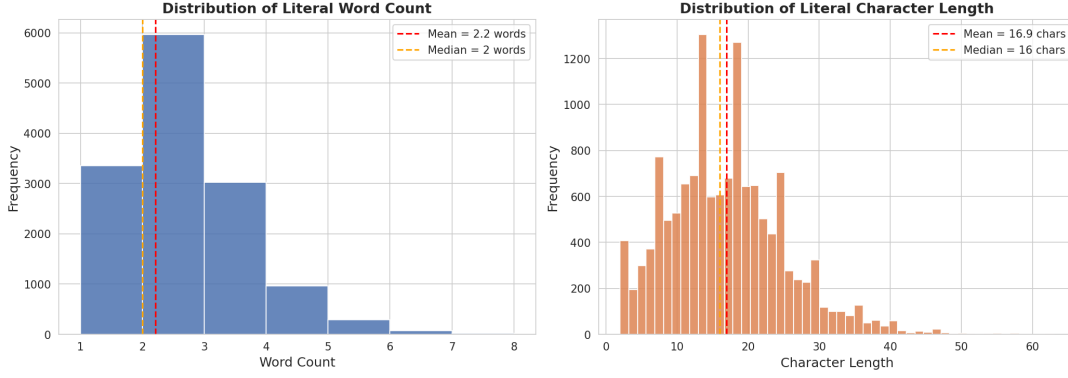


Figure 2: Distribution of Text Lengths

### 3 Methodology

To establish performance bounds, we evaluated two distinct architectural paradigms. Both were trained and tested on the identical 80/20 stratified split (using random seed 42) to ensure a rigorous comparison.

#### 3.1 Baseline Pipeline (TF-IDF + Logistic Regression)

We built a baseline system in Scikit-Learn utilizing a hybrid feature extraction pipeline:

1. **Character  $n$ -gram Vectorizer:** Extracts character  $n$ -grams ranging from 3 to 5 characters (maximum features: 100,000). This captures clinical root words, suffixes, prefixes, and minor typos.
2. **Word  $n$ -gram Vectorizer:** Extracts unigrams and bigrams (maximum features: 50,000) to represent whole terms.

These vectorizer outputs are joined via a `FeatureUnion` and passed to a multinomial Logistic Regression classifier with LBFGS optimization and default regularization ( $C = 1.0$ ).

#### 3.2 Deep Learning Pipeline (RoBERTa + Attention Pooling)

The neural network model is built on top of the pre-trained clinical language model `PlanTL-G0B-ES/roberta-base` which has been pre-trained on Spanish clinical texts. To capture textual features effectively, we discarded the simple class-token representation and implemented:

- **Attention Pooling:** Calculates a weighted sum over the token embeddings using a learnable attention mechanism, allowing the model to focus on key clinical terms.
- **Multi-Sample Dropout:** We used 5 parallel dropout layers ( $p = 0.1$ ) before the final classification head to reduce overfitting and speed up convergence.
- **Loss Optimization:** The model was trained using Cross-Entropy Loss with the AdamW optimizer, a cosine learning rate scheduler with warmup, and gradient accumulation.

## 4 Quantitative Results

The evaluation on the 2,740 validation samples shows that the deep learning model outperforms the TF-IDF baseline across all core metrics. The comparison is summarized in Table 1.

Table 1: Model Evaluation and Comparison (Validation Set)

| Metric          | TF-IDF + Logistic Regression | RoBERTa + Attention Pooling | Delta ( $\Delta$ ) |
|-----------------|------------------------------|-----------------------------|--------------------|
| Accuracy        | 52.99%                       | <b>57.01%</b>               | <b>+4.02%</b>      |
| Macro- $F_1$    | 44.49%                       | <b>49.42%</b>               | <b>+4.93%</b>      |
| Weighted- $F_1$ | 51.04%                       | <b>55.07%</b>               | <b>+4.03%</b>      |

The absolute improvement in Macro- $F_1$  (+4.93%) is particularly notable because it indicates that the deep learning model handles class imbalance more robustly, achieving better recall and precision on the tail categories compared to the baseline.

## 5 Detailed Error Analysis

An error analysis was performed on the validation set predictions to characterize the deep learning model’s failure modes. Out of 2,740 validation samples, the deep learning model committed **1,178 errors** (a 43.0% error rate). The confusion heatmap is visualized in Figure 3, and the top 15 most frequent error pairs are plotted in Figure 4.

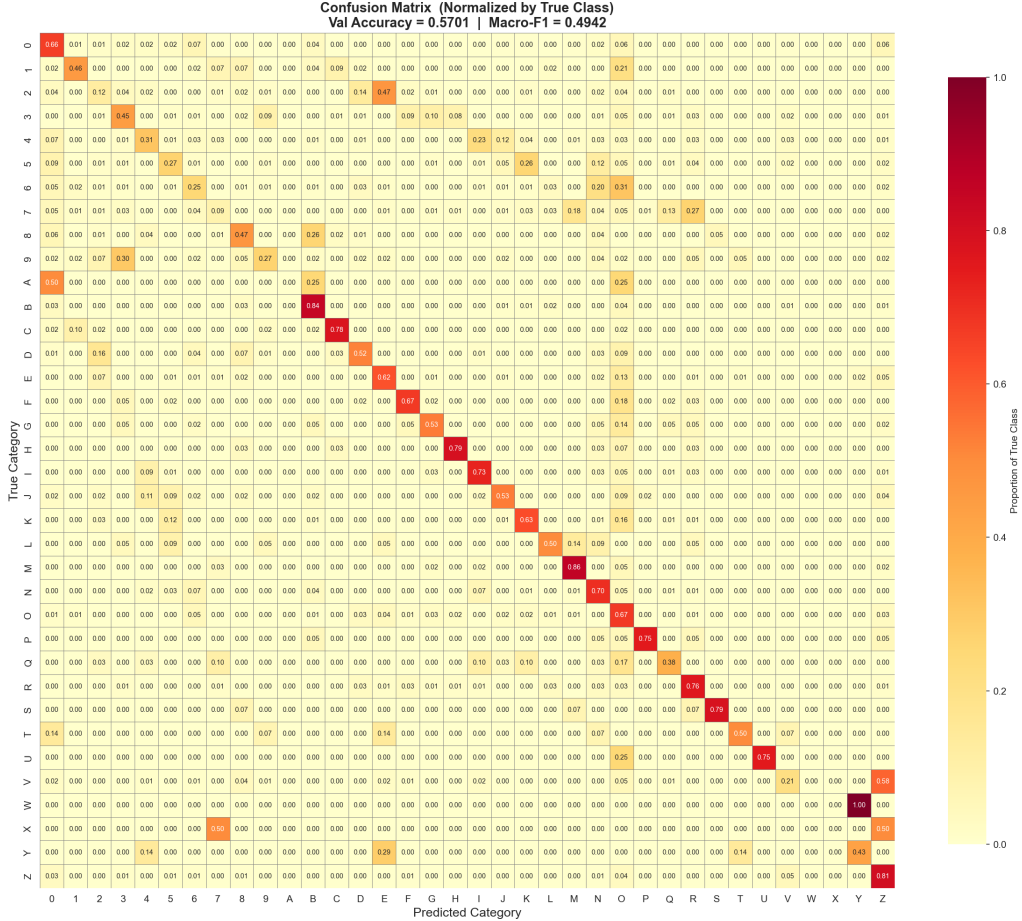


Figure 3: Normalized Confusion Heatmap

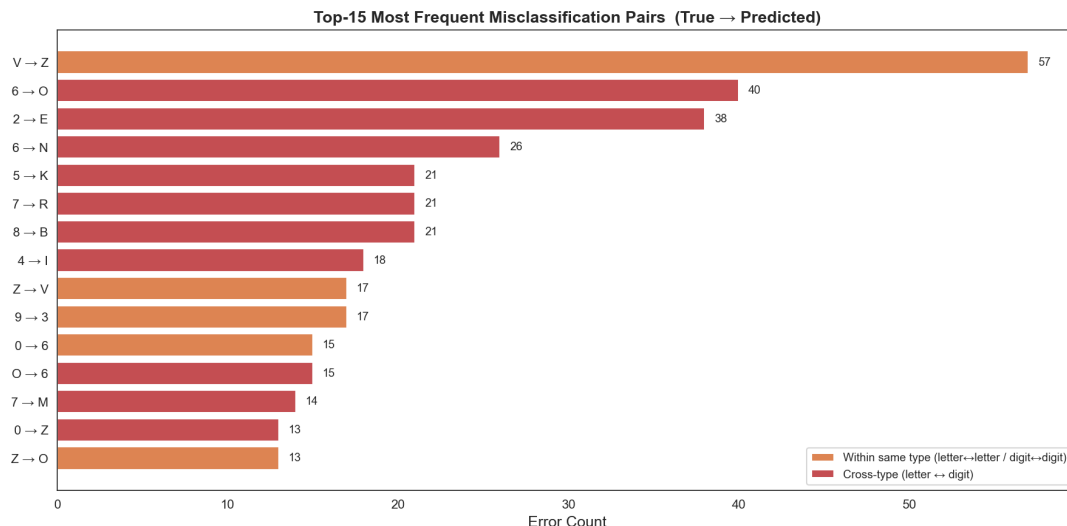


Figure 4: Top 15 Misclassification Pairs

## 5.1 Analysis of Specific Error Modes

### 5.1.1 Neoplasm Subtype Confusion (Chapter C vs. D)

We investigated the confusion between Malignant Neoplasms (C) and Blood/Benign Neoplasms (D). Surprisingly, the model displays exceptional granularity, making only **2 errors** total between these chapters on the validation set.

- *Literal*: “Adenocarcinoma in situ endocervical” → True: D, Pred: C (Confidence: 0.600)
- *Literal*: “tumor mucinoso borderline ovario izq” → True: D, Pred: C (Confidence: 0.629)

This indicates that the clinical BERT embedding space is well-attuned to the semantic difference between benign/borderline and malignant terms (such as “in situ” or “borderline”).

### 5.1.2 Letter and Digit Look-Alikes

Due to the alphanumeric layout, we analyzed whether standard characters are confused with similar-looking digits:

- **Letter O (Pregnancy) vs. Digit 0 (Procedures)**: Only **4 errors** occurred. These are mostly related to injuries or childbirth complications that the model maps to procedure-based words (e.g., “Desgarro 1” → True: O, Predicted: 0, Confidence: 0.708).
- **Letter Z (Health Status) vs. Digit 0 (Procedures)**: **12 errors**. This confusion stems from surgical history described as a current status (e.g., “Anexectomía izquierda”, “Histerectomía”, “By pass Aortobifemoral”), where the model maps the surgical nouns to surgical procedures (0) rather than health-status codes (Z).
- **Digit 6 (Extracorporeal Procedures) vs. Letter O (Pregnancy)**: This is a major confusion axis, representing **40 errors** where true 6 was predicted as O, and **15 errors** where true O was predicted as 6. This is driven by obstetric monitoring procedures (e.g., Non-Stress Tests or labor tracking) occurring during birth (e.g., “itu pARTO” → True: 6, Predicted: O, Confidence: 0.809).

### 5.1.3 Top Global Confusions

The most frequent error pair in the entire validation set is **V (Transport accidents) predicted as Z (Health status / Services)** with 57 errors. This occurs because patients presenting with injuries or histories of accidents are often documented in ways that resemble routine checkups or follow-up consults, leading the model to predict the general clinic visit code (Z) instead of the external injury cause (V). An example includes:

- *Literal*: “alergia Àcars” (dust mite allergy) → True: V (likely a labeling anomaly), Predicted: Z (Confidence: 0.990)

## 5.2 High-Confidence Errors

There are 402 instances where the model predicted a category with  $\geq 70\%$  confidence but was incorrect. A qualitative examination reveals that many of these errors are due to **annotation noise** or highly ambiguous literals in the dataset:

- *Literal*: “Febre rn” (Newborn fever) → True: 7 (Osteopathic Procedures - likely an administrative coding error), Predicted: P (Perinatal Conditions, Confidence: 0.991)
- *Literal*: “Cataratas bilateras” (Bilateral cataracts) → True: 3 (Measurement/Administration), Predicted: H (Diseases of the Eye, Confidence: 0.980)
- *Literal*: “Hipoacusia bilateral” (Bilateral hearing loss) → True: 3 (Measurement/Administration), Predicted: H (Diseases of the Ear, Confidence: 0.975)

These cases indicate that the model’s errors are often semantically logical, and that the ground-truth annotations in the clinical database contain systematic noise or specialized administrative guidelines that do not align with raw text semantics.

## 6 Conclusion and Future Work

This report outlines the successful implementation and evaluation of a clinical language model for Spanish ICD-10 chapter classification. By moving from a TF-IDF baseline to a pre-trained RoBERTa model with attention pooling, we achieved a **+4.02% Accuracy** and **+4.93% Macro- $F_1$**  improvement.

Future iterations of the pipeline should focus on:

1. **Imbalance Mitigation:** Applying loss-weighting techniques (e.g., Focal Loss or class-balanced weights) to improve performance on tail classes with under 20 training samples.
2. **Data Cleaning:** Conducting manual review and filtering of high-confidence errors to correct systematic administrative misannotations (such as mapping cataracts to administrative codes rather than eye disorders).
3. **Tokenization Adjustments:** Injecting character-level representations into the transformer embeddings to help resolve look-alike numeric vs. alphabetic ambiguities.